

CS 217: Practice questions on Generative Classifiers

- Consider a binary classification problem ($y \in \{0, 1\}$) and two dimension data ($d = 2$) where both x_1 and x_2 are binary. Let $n_{00}^y, n_{10}^y, n_{01}^y, n_{11}^y$ denote the number of training instances in class y where $[x_1 \ x_2]$ is $[0 \ 0]$, $[1 \ 0]$, $[0 \ 1]$, $[1 \ 1]$ respectively.

(a) Suppose we use a naive Bayes classifier where x_1 is assumed independent of x_2 .

- What is the maximum likelihood estimate of the two Bernoulli parameters in class y in terms of the counts $n_{00}^y, n_{10}^y, n_{01}^y, n_{11}^y$..1

$$p^y(x_1 = a) = \frac{n_{a0}^y + n_{a1}^y}{n_{00}^y + n_{10}^y + n_{01}^y + n_{11}^y} \quad p^y(x_2 = a) = \frac{n_{0a}^y + n_{1a}^y}{n_{00}^y + n_{10}^y + n_{01}^y + n_{11}^y}$$

- What is the maximum likelihood estimate of $\Pr(y)$ in terms of the counts $n_{00}^y, n_{10}^y, n_{01}^y, n_{11}^y$..1

$$p(y = 0) = \frac{n_{00}^0 + n_{10}^0 + n_{01}^0 + n_{11}^0}{\sum_{a \in \{0,1\}} n_{00}^a + n_{10}^a + n_{01}^a + n_{11}^a}$$

- For what values of $n_{00}^y, n_{10}^y, n_{01}^y, n_{11}^y$ will the naive Bayes classifier incur the maximum percentage training error? Justify. Assume number of instances in both classes is the same. ..3

If $n_{00}^0 = n_{11}^0, n_{10}^0 = n_{01}^0 = 0$ and $n_{01}^1 = n_{10}^1, n_{00}^1 = n_{11}^1 = 0$, the estimated Bernoulli parameter will be $1/2$ in both classes. The $\Pr(\mathbf{x}|y)$ value will also be the same. The error will be 50%.

- Now suppose we decide to use a model more powerful than naive Bayes. In each class, we model the probability that $\Pr([x_1 \ x_2] = [a \ b] | y)$ with a parameter p_{ab}^y . Further we share some parameters between the two classes to reduce the number of parameters by requiring that $p_{11}^0 = p_{11}^1 = p_{11}$. Clearly, $p_{00}^y + p_{10}^y + p_{01}^y + p_{11} = 1$. Write the log-likelihood function on the training data in terms of the counts n_{ab}^y , and parameters p_{ab}^y , $\Pr(y = 1), \Pr(y = 0)$. ..2

$$\begin{aligned} \sum_i \log P(y_i) * P(y_i|x) &= \sum_i \log P(y_i) + \sum_i \log P(y_i|x) \\ &= n^1 * \log P(y = 1) + n^0 * \log P(y = 0) + \sum_y (n_{00}^y * \log p_{00}^y + n_{01}^y * \log p_{01}^y + n_{10}^y * \log p_{10}^y) + \\ &\quad (n_{11}^0 + n_{11}^1) * \log P_{11} \end{aligned}$$

- Derive the maximum likelihood estimates of the $p_{00}^y, p_{10}^y, p_{01}^y$ parameters only in terms of p_{11} and the training counts. [Hint: Use Lagrangian multiplier to push the two constraints $p_{00}^y + p_{10}^y + p_{01}^y + p_{11} = 1$, one for each y to the objective.] ..3

We Maximize the objective created above (F_{ab}^y) using the lagrangian dual.

We thus have the objective as: Maximize $F_{ab}^y - \lambda_1 * (p_{00}^y + p_{10}^y + p_{01}^y + p_{11}) - \lambda_0 * (p_{00}^0 + p_{01}^0 + p_{10}^0 + p_{11}^0)$

$$\frac{\partial F(P_{ab}^y)}{\partial P_{ab}^y} = \frac{n_{ab}^y}{P_{ab}^y} - \lambda_y$$

Using the constraints we get $\frac{n_{00}^y}{\lambda_y} + \frac{n_{01}^y}{\lambda_y} + \frac{n_{10}^y}{\lambda_y} + P_{11} = 1$ Thus $P_{ab}^y = \frac{n_{ab}^y}{(n^y - n_{11}^y)} * (1 - P_{11})$

- Now derive the maximum likelihood estimate for p_{11} . ..1

Using the values obtained above in $F(P_{ab}^y)$ and taking only the terms concerning P_{11} we get $\log(1 - P_{11}) * \sum_y (n_{00}^y + n_{10}^y + n_{01}^y) + (n_{11}^0 + n_{11}^1) * \log P_{11}$

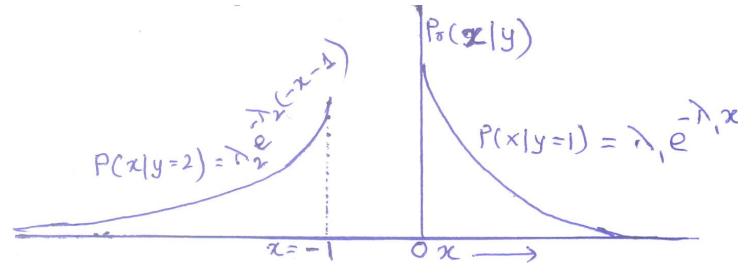
On differentiating this with P_{11} and solving for P_{11} we get $P_{11} = \frac{(n_{11}^0 + n_{11}^1)}{(n^0 + n^1)}$

- The exponential density function is a popular parametric density function to capture the distribution of positive reals z . It is defined as:

$$\begin{aligned} \exp(z; \lambda) &= \lambda e^{-\lambda z} \quad \text{if } z \geq 0 \\ &= 0 \quad \text{otherwise.} \end{aligned}$$

The mean, variance, cumulative density of the above are $\lambda^{-1}, \lambda^{-2}, 1 - e^{-\lambda z}$ respectively.

Suppose you apply the naive Bayes classifier on a dataset where the *true* $\Pr(x|y)$ distribution is defined as $\Pr(x|y = 1) = \exp(x; \lambda_1)$ and $\Pr(x|y = 2) = \exp(-1 - x; \lambda_2)$. Assume $\lambda_1 > 0, \lambda_2 > 0$. The picture below shows an example distribution for the two classes with different λ_1, λ_2 .



The true values of λ_1, λ_2 is not known to us, nor do we know that the data was generated from this special density function. We are only given a training set D by sampling a large number $N/2$ of instances from each of the two classes. When training the naive Bayes classifier, we will continue to use a Gaussian distribution to model $P(x_1|y)$ for each of the two classes. Thus, we will need to train parameters mean μ_y and variance σ_y^2 for each y .

- (a) Write down the maximum likelihood estimates of the following parameters (in terms of λ_1, λ_2) assuming N is very large. The first question is solved for you to give you an idea of the solution:

i. μ_1

Answer: λ_1^{-1}

ii. μ_2

$-\lambda_2^{-1} - 1$

..1

iii. σ_1^2 , and σ_2^2

(λ_1^{-2}) and (λ_2^{-2}) respectively.

..2

- (b) For what value of w_2, w_1, w_0 is predicting class 1 based on the hyperplane $w_2 x^2 + w_1 x + w_0 > 0$, the same as the predictions made by the above naive Bayes classifier.

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Our class priors are equal.

$$\begin{aligned} \log \Pr(y = 1|x) &\geq \log \Pr(y = 2|x) \\ \rightarrow \frac{-(x_1 - \mu_1)^2}{2 * \sigma_1^2} - \log \sigma_1 &\geq \frac{-(x_1 - \mu_2)^2}{2 * \sigma_2^2} - \log \sigma_2 \end{aligned}$$

Substitute in the above the value of $\mu_1 = \lambda_1^{-1}$, $\mu_2 = -\lambda_2^{-1} - 1$, and $\sigma_y = \lambda_y^{-2}$. Double check if that will give us:

$$w_2 = -\lambda_1^2 + \lambda_2^2, w_1 = 2(\lambda_1 + \lambda_2(\lambda_2 + 1)), w_0 = \log \lambda_1 - \log \lambda_2 + (\lambda_2 + 1)^2 - 1$$

- (c) For what value of λ_1, λ_2 is the training error minimum? What is the value of the minimum error? Justify.

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The training error will be zero for any $\lambda_1 = \lambda_2$